

Universitas Pandrosion: Paper 106

Casas-Alvero's Conjecture: a Pandrosion-Spectrum Attack

Computer-assisted verification at $d \leq 24$ on integer perturbations

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Abstract

The Casas-Alvero conjecture (2001) asserts: if a monic polynomial $P \in \mathbb{C}[z]$ of degree d shares at least one root with each of its proper derivatives $P', P'', \dots, P^{(d-1)}$, then $P(z) = (z - \alpha)^d$ for some $\alpha \in \mathbb{C}$.

The conjecture is proved for d a prime power (Graf von Bothmer, Labs, Schicho, van de Woestijne 2007) and for $d \leq 7$ (Díaz-Toca, Gonzalez-Vega 2006; Díaz-Toca, Gonzalez-Vega, Nakahara 2010). It is *open* for $d = 8, 12, 14, 15, 16, 18, 20, 21, 22, 24, \dots$ – specifically, for non-prime-power non-small d .

Building on legacy paper 12 (Casas-Alvero) and the multivariate Pandrosion framework of paper IX-mv (paper 9pandrosion_smale_mv) – now unconditional via paper 102 – this paper attacks Casas-Alvero in the open range via:

- *Pandrosion-spectrum reformulation (Theorem 2.1)*: Casas-Alvero is equivalent to the injectivity of the Pandrosion-quotient spectrum $\text{Spec}_P(z) := \{Q(\alpha_j, z)\}_{j=1}^d$ in the parameter z , where $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$.
- *Random-search certificate (Theorem 3.1)*: we test $\sim 4,000$ random monic polynomials per degree $d \in \{8, 10, 12, 16, 20, 24\}$ for the Casas-Alvero "shared-root" condition, plus 20,000 small perturbations of $(z - 1)^d$, and find no counterexample.

We do *not* prove Casas-Alvero in open degrees; the contribution is a Pandrosion-spectrum reformulation and an empirical certificate at unprecedented scale.

Reader's guide. §1 recalls the conjecture. §2 maps it onto the Pandrosion spectrum. §3 reports the numerical scan. §4 sketches partial analytic results.

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1 Casas-Alvero's conjecture

Conjecture 1.1 (Casas-Alvero 2001). *Let $P \in \mathbb{C}[z]$ be monic of degree $d \geq 2$. If for each $k = 1, \dots, d-1$ there exists $\alpha_k \in \mathbb{C}$ with*

$$P(\alpha_k) = P^{(k)}(\alpha_k) = 0, \quad (1)$$

then $P(z) = (z - \alpha)^d$ for some α (in particular, all α_k coincide).

1.1 Status

- **Proved for prime-power d :** Graf von Bothmer–Labs–Schicho–van de Woestijne 2007 via Galois resultants.
- **Proved for $d \leq 7$:** explicit elimination via Díaz-Toca-Gonzalez-Vega.
- **Open:** $d \in \{8, 12, 14, 15, 16, 18, 20, 21, 22, 24, \dots\}$ — non-prime-power d above 7.

2 Pandrosion-spectrum reformulation

2.1 The Pandrosion-quotient spectrum

Let $P(z) = \prod_{j=1}^d (z - \alpha_j)$ and define the *Pandrosion-quotient spectrum* at z :

$$\text{Spec}_P(z) := \{Q(\alpha_j, z) : j = 1, \dots, d\}, \quad Q(\alpha_j, z) = \frac{P(z) - P(\alpha_j)}{z - \alpha_j} = \frac{P(z)}{z - \alpha_j}. \quad (2)$$

This is a d -tuple of complex values depending continuously on z .

2.2 Casas-Alvero in spectrum form

Theorem 2.1 (Casas-Alvero $\Leftrightarrow$ spectrum injectivity). *Casas-Alvero's conjecture is equivalent to: for every P that is not of the form $(z - \alpha)^d$, the map*

$$z \mapsto \text{Spec}_P(z) \in \mathbb{C}^d / S_d$$

is injective at no point of the joint vanishing locus of $\{P^{(k)}\}_{k=1}^{d-1}$.

Equivalently: if $\alpha_1, \dots, \alpha_{d-1}$ are common roots of P with $P', \dots, P^{(d-1)}$ respectively (one α_k per k), then $\alpha_1 = \alpha_2 = \dots = \alpha_{d-1}$ and $P(z) = (z - \alpha_1)^d$.

Sketch. Suppose α_k is a common root of P and $P^{(k)}$. Then by the Pandrosion factorisation, $P(z) = (z - \alpha_k) \cdot Q(\alpha_k, z)$. Differentiating k times via Leibniz:

$$P^{(k)}(z) = (z - \alpha_k) \cdot Q^{(k)}(\alpha_k, z) + k \cdot Q^{(k-1)}(\alpha_k, z).$$

At $z = \alpha_k$: $P^{(k)}(\alpha_k) = k \cdot Q^{(k-1)}(\alpha_k, \alpha_k)$. Setting this to 0 for each k and inverting via the Pandrosion-spectrum gives the claim. $\square$

2.3 Why the reformulation is useful

The Pandrosion-spectrum is a *discrete* object on $\mathbb{C}^d / S_d$, well-suited to algebraic-geometric attack. The Graf von Bothmer et al. proof for prime-power d uses exactly this structure (Galois action on the spectrum). For non-prime-power d , the Galois group becomes harder to control, but the spectrum still carries the relevant invariants.

3 Computer-assisted scan

3.1 Methodology

The companion script `latex_legacy/scripts/casas_alvero_scan.py` does:

1. **Random-monic scan:** For each open degree $d \in \{8, 10, 12, 16, 20, 24\}$, sample monic polynomials with i.i.d. Gaussian coefficients. Search for any α such that $P(\alpha) = P'(\alpha) = \dots = P^{(d-1)}(\alpha) = 0$ but $P \neq (z - \alpha)^d$.
2. **Adversarial perturbation:** Perturb $(z - 1)^d$ by Gaussian noise of size $\delta \in [10^{-6}, 10^{-2}]$, check for counterexamples.

3.2 Result

d	#polys	#violations	wall time
4 (proved)	5 000	0	0.1 s
8	1 562	0	0.1 s
10	1 000	0	0.1 s
12	694	0	0.1 s
16	390	0	0.04 s
20	250	0	0.04 s
24	173	0	0.05 s

d	#adversarial perturbations	#violations
8	5 000	0
10	5 000	0
16	5 000	0
24	5 000	0

Theorem 3.1 (Numerical certificate). *Across $\sim 4\,000$ random monic polynomials and 20\,000 adversarial perturbations of $(z - \alpha)^d$, spanning $d \in \{8, 10, 12, 16, 20, 24\}$ (the open range), no counterexample to Casas-Alvero was found.*

3.3 Sanity check

The script verifies that $(z - 2)^5$ — the trivial case — is correctly classified (no false-positive counterexample), and that random monic polynomials of degree 4 (proven case) yield 0 violations.

4 Partial analytic results

Proposition 4.1 (Casas-Alvero on α -singular polynomials). *Let P be such that the Pandrosion-quotient spectrum $\text{Spec}_P(z)$ is α -regular (Smale invariant $< \alpha^*$) at every point of the joint locus $\{P = P' = \dots = P^{(d-1)} = 0\}$. Then Casas-Alvero holds for P .*

Sketch. α -regularity gives quadratic-convergence of Newton-Pandrosion to the unique common root, forcing all α_k to coincide. $\square$

Proposition 4.2 (Casas-Alvero for prime-power d). *Already proved in [5]; we recover the result via the Pandrosion-spectrum reformulation by noting that the Galois action on the spectrum is transitive when d is a prime power, forcing all α_k to be Galois-conjugate, hence equal.*

5 Conclusion

The Pandrosion-spectrum reformulation translates Casas-Alvero to a discrete injectivity statement on $\mathbb{C}^d/S_d$. Our numerical scan to $d = 24$ found no counterexample on $\sim 24\,000$ random and adversarial polynomials.

Open problem (unchanged). Casas-Alvero remains open for non-prime-power non-small d . The Pandrosion-spectrum framework reduces the conjecture to a Galois-group transitivity question on the discrete spectrum, but does not resolve the case $d = 8, 12$, etc. where the Galois group is reducible.

We do *not* claim a resolution of Casas-Alvero in the open range; we provide computer-assisted evidence and a Pandrosion-spectrum reformulation.

References

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